

# AKANKSH GATLA

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## PROFESSIONAL SUMMARY

Passionate Software Development Engineer (Data Engineering) with 4+ years of expertise architecting and deploying enterprise-scale distributed data platforms across fintech and healthcare domains. Deep technical proficiency in Apache Spark, PySpark, Databricks, Snowflake, AWS Glue, and Azure Data Factory ecosystems. Proven ability to design and optimize petabyte-scale ETL/ELT pipelines, implement machine learning workflows for predictive analytics, and drive AI-powered automation initiatives. Expert in building cloud-native data architectures on AWS and Azure, leveraging advanced optimization techniques including partitioning, clustering, and indexing to achieve significant performance improvements. Strong track record of reducing processing latency by up to 40%, implementing robust CI/CD frameworks, and delivering analytics-ready datasets that enable data-driven strategic decisions across organizations.

## PROFESSIONAL EXPERIENCE

### ARTECH LLC, CLIENT: CAPITAL ONE

*App Dev - Java - Sr.*

*Plano, TX*

Nov 2025 – Present

- Built a new ETL feature using Databricks operational compute, sourcing Snowflake views and S3 Delta datasets, with Airflow-triggered orchestration for daily TB-scale runs.
- Designed region-agnostic configuration generation from YAML, producing in-memory, region-specific runtime configs for Glue and Databricks jobs across regions.
- Delivered end-to-end pipeline rebuild for Business Bureau source format changes, handling schema/version shifts and downstream logic updates with 100% parity in validation outputs.
- Implemented data validation and reconciliation checks across multi-source inputs to reduce load failures by 30-40% in the first month.
- Optimized Spark job performance through partitioning and broadcast joins, cutting runtimes by 20-30% and stabilizing peak-hour SLAs at 99.5%+ on-time.
- Aligned data contracts with cross-functional teams, reducing integration rework by 25% and accelerating change approval cycles.
- Automated alerting and runbook-driven remediation for ETL failures, improving MTTR by 35-45% and shrinking on-call escalations.

**Technologies:** Databricks, Apache Spark, PySpark, Airflow, Snowflake, AWS S3, Delta Lake, AWS Glue, YAML, SQL, Python

### GENZEON CORPORATION

*Data Integration Engineer*

*Exton, PA*

Aug 2025 – Nov 2025

- Architected end-to-end ingestion pipelines processing HL7 feeds and EPIC FHIR databases into Azure Data Lake using Spark and Delta Lake for 350+ healthcare clients.
- Built intelligent automation framework in Databricks with PySpark and ML algorithms, reducing manual data mapping effort by 40% and improving data quality.
- Designed Medallion Architecture (Bronze/Silver/Gold) Databricks Jobs delivering curated views that improved downstream analytics performance by 30%.

**Technologies:** Azure Databricks, PySpark, Apache Spark, ADLS Gen2, Delta Lake, HL7, FHIR, Airflow, SQL, Python, Medallion Architecture

### D-VITAL LLC, CLIENT: CAPITAL ONE

*Data Engineer*

*Dublin, OH*

Nov 2024 – Aug 2025

- Engineered Standard Performance Reporting application integrating OneLake with AWS S3, orchestrating 20+ Glue Spark ETL jobs through Step Functions for automated daily processing.
- Implemented CI/CD pipelines with GitHub Actions, Jenkins, and Terraform, cutting deployment time by 40% using Blue-Green strategies for zero-downtime releases.
- Optimized Snowflake warehouse performance by 40% through query profiling, reengineering complex joins, and implementing materialized views for heavily accessed datasets.
- Enhanced query performance by 30% by architecting Snowflake Tables with compound indexing and micro-partitioning on 1.6M+ row datasets.
- Led cross-functional teams to build AWS Serverless Data Products (Lambda, Step Functions, S3, Glue), delivering MVP within 6-week sprints.
- Integrated Credit Bureau data feeds (Equifax, TransUnion, Experian) into 1600+ field schema using PySpark, applying SCD2/SCD1 for historical tracking.
- Strengthened security posture by remediating vulnerabilities, implementing RBAC IAM policies, and developing test frameworks achieving 95% coverage with pytest.

**Technologies:** AWS (S3, Glue, Step Functions, Lambda, Athena, Redshift), Snowflake, Databricks, PySpark, Terraform, Jenkins, GitHub Actions, pytest, behave

### STUEBS ENTERPRISE INC, CLIENT: CODE NINJAS

*Sensei/Coding Instructor*

*Dublin, OH*

Sep 2024 – Oct 2024

- Instructed 30+ students aged 7-14 in programming fundamentals, game development, and robotics, delivering hands-on coding curriculum across JavaScript, Python, and Scratch.
- Facilitated STEM workshops and coding camps, mentoring students in building real-world projects including mobile games, web applications, and Roblox creations.

**Technologies:** JavaScript, Python, Scratch, Roblox Studio, Lua, Game Development, STEM Education

### DATALYSYS LLC

*Intern Data Engineer*

*Dublin, OH*

Feb 2024 – Oct 2024

- Architected HIPAA-compliant ELT platform using Azure Data Factory and Databricks, automating 8-hourly EMR data extraction via RESTful APIs with OAuth 2.0.
- Implemented Medallion Architecture with PySpark and Delta Lake, applying SCD2 patterns and CDC for real-time clinical analytics and regulatory reporting.
- Built ML-powered automation system generating non-compliant patient cohorts for Value-Based Care, integrating with EMRs via HL7/FHIR APIs.

- Developed UDS reporting platform with Spark SQL and Power BI for HEDIS compliance, optimizing jobs through DataFrame caching and broadcast joins.
- Deployed applications with CI/CD pipelines via Jenkins, reducing deployment time by 30% while achieving 95% test coverage.

**Technologies:** Azure Data Factory, Databricks, PySpark, ADLS, Delta Lake, Power BI, DAX, Spark SQL, OAuth 2.0, HL7/FHIR, Jenkins, pytest

**DATALYSYS SOFTWARE SOLUTIONS PVT LTD**  
**Associate Data Engineer**

**Remote**  
Sep 2020 – Jul 2022

- Orchestrated complex ETL workflows using Airflow DAGs automating SQL Server processes with dynamic pipeline generation and SLA monitoring.
- Integrated OAuth 2.0 for EMR API access and conducted Risk Stratification Analysis using K-Means and Hierarchical clustering on clinical data.
- Developed PySpark analytics workflows in Databricks processing 3.8M patient records with partitioning and broadcast joins for predictive risk models.
- Architected AI-powered Patient Chatbot with NLP on Azure Bot Service, handling 5K+ monthly interactions with 92% satisfaction rate.
- Implemented feature engineering pipelines (Scaling, PCA, K-Means, DBSCAN) enhancing patient risk stratification accuracy by 35%.
- Automated Clustering Analysis for 350+ ICD-10 codes with Python and PySpark, creating Tableau dashboards for clinical decision support.
- Managed agile workflows using Jira and MS Teams for sprint planning and cross-functional collaboration.

**Technologies:** Apache Airflow, PySpark, Databricks, Azure Bot Service, Cognitive Services, SQL Server, Python, scikit-learn, NLP, Tableau, Jira

**TECHNICAL SKILLS**

|                                           |                                                                                                                                          |
|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
| Languages                                 | Python 3.x, R, SQL, T-SQL, Spark SQL, PySpark, Java                                                                                      |
| Database                                  | PostgreSQL, SQL Server, MSSQL, MySQL, SQLite, Snowflake, MongoDB                                                                         |
| Methodologies                             | Agile, Scrum, Waterfall                                                                                                                  |
| Operating Systems                         | Linux (Any Distro), Unix, Windows                                                                                                        |
| Cloud Services                            | AWS (S3, EC2, Lambda, Glue, Step Functions, Athena, Redshift), Azure (Data Factory, Databricks, ADLS, Bot Service), GCP                  |
| Big Data Framework                        | Apache Hadoop, Spark, Kafka, Solr, Pig, Hive, Databricks, Snowflake                                                                      |
| Data Integration                          | ETL/ELT, Azure Data Factory, AWS Glue, Step Functions, Airflow, Erwin Modelling                                                          |
| Statistical Methods                       | Hypothesis Testing, ANOVA, Time Series, Confidence Intervals, Bayes Law, PCA, Dimensionality Reduction, Cross-Validation                 |
| Business Intelligence & Predictive Models | Regression Analysis, Bayesian Method, Decision Tree, Random Forests, SVM, Neural Network, K-Means Clustering, KNN, Ensemble Methods, NLP |
| Reporting Tools                           | Tableau (Desktop, Server, Online), Microsoft Power BI, Smartsheet, Google Sheets, Google Data Studio, Microsoft Excel                    |
| Data Visualization                        | Tableau, Microsoft Power BI, Matplotlib, Seaborn, Plotly, Microsoft Excel                                                                |
| Machine Learning                          | Regression, Clustering, SVM, Decision Trees, Classification, Recommendation Systems                                                      |
| ETL/Data Warehouse                        | Talend, Informatica, Data Modeling (Star Schema, Snowflake Schema), FACT and Dimension Tables, Pivot Tables                              |
| Web Framework                             | Flask, Django, Streamlit                                                                                                                 |

**EDUCATION**

**UNIVERSITY AT BUFFALO**  
**Master of Science in Computer Science**

**Buffalo, New York**

Specialized in Big Data Analytics, Distributed Systems, and Machine Learning

**LOVELY PROFESSIONAL UNIVERSITY**  
**Bachelor of Technology in Computer Science & Engineering**

**Punjab, India**

Specialized in Data Structures, Algorithms, Database Systems, and Cloud Computing

**ACADEMIC PROJECTS**

**GLOBAL SUPER STORE**

Automated ETL pipeline using AWS Lambda and Python to process Super Store API data, reducing manual data processing time by 60%, enabling real-time analytics and insights through Amazon S3, AWS Glue, and Athena for streaming data in the cloud.

**HEALTH CARE APPLICATION**

Engineered Healthcare prediction system with 89% F1 score with Random Forest Classifier, deployed on AWS EC2, via Putty featuring an interactive Flask based front end integrated with Tableau Dashboard's for patients.

Published: "Predictions And Analytics in Healthcare: Advancements in Machine Learning" IRJET Machine Learning for Healthcare Workshop (MLHC).

**CERTIFICATIONS**

- Databricks Certified Data Engineer Associate
- Microsoft Certified: Azure Data Engineer Associate
- Databricks Certified: Generative AI Fundamentals
- Google Certified: Introduction to Generative AI
- Jovian Certified: Data Analysis with Python : Zero to Pandas